



(Approved by AICTE, New Delhi & Affiliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173



SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (2026–
2027)

PROGRAM BOOK FOR
SHORT-TERM INTERNSHIP

Name of the Student: JONNADA MANGA

Registration Number: 323129510025

Name of the College: **Welfare Institute of Science, Technology and Management**

Period of Internship: From: 21-05-2026 To: 21-07-2026

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

SENTIMENT ANALYSIS SYSTEM FOR MOVIE REVIEWS USING NATURAL LANGUAGE PROCESSING TECHNIQUES

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs. K. Puspha Latha

Department of CSE

Welfare Institute of Science, Technology and Management

Submitted by:

JONNADA MANGA

Reg.No: 323129510025

Department of CSE

Department of COMPUTER SCIENCE ENGINEERING

Welfare Institute of Science, Technology and Management

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Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - a. Data and information you are expected to collect about the organization and/or industry.
 - b. Job skills you are expected to acquire.
 - c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your college.

Student's Declaration

I, **JONNADA MANGA**, a student of **Bachelor of Technology** Program, Reg. No. **323129510025** of the Department of **Computer Science Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.K.Puspha Latha** Department of **Computer Science Engineering, Wellfare Institute of Science, Technology and Management.**

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature reads "J. Manga" in a cursive, slightly stylized font. Below the name, there are some faint, illegible markings that appear to be a date or other text.

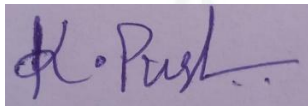
(Signature and Date)

Official Certification

This is to certify that **JONNADA MANGA** , Reg.No. **323129510025** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **SENTIMENT ANALYSIS SYSTEM FOR MOVIE REVIEWS USING NATURAL LANGUAGE PROCESSING TECHNIQUES** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science Engineering** at **Welfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department
Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **JONNADA MANGA**, Reg.No. **323129510025** of **Welfare Institute of Science, Technology and Management**, underwent internship in **SENTIMENT ANALYSIS SYSTEM FOR MOVIE REVIEWS USING NATURAL LANGUAGE PROCESSING TECHNIQUES** And Sentiment Analysis For Social Media Applications at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Welfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs. Nalla Suneetha**, Head of the Department of **Computer Science Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.K.Puspha Latha** Assistant Professor of the Department of **Computer Science Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my 8-week Short-Term Internship in Sentiment Analysis System for Movie Reviews Using Natural Language Processing Techniques, conducted at the Council for Skills and Competencies (CSC India). The internship spanned from 1-05-2025 to 30-06-2025 and was undertaken as part of the academic curriculum for the Bachelor of Technology. The primary objective of this internship was to gain proficiency in Natural Language Processing (NLP), Machine Learning, data analysis, and reporting to enhance employability skills.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a sentiment analysis system using Python, Natural Language Processing (NLP), and machine learning libraries (scikit-learn, pandas) that can classify movie reviews.
- To integrate text preprocessing techniques for understanding user reviews, including tokenization, stop-word removal, and TF-IDF vectorization for feature extraction.
- To implement and compare multiple classification algorithms including Naive Bayes, Logistic Regression, and Random Forest to determine the most effective approach for sentiment analysis.
- To create comprehensive data visualizations that make sentiment trends, rating distributions, and model performance metrics accessible and understandable.
- To enable the system to act as an analytical tool by generating actionable insights from large volumes of unstructured textual data, thereby improving decision-making for movie platforms.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational sentiment analysis system capable of understanding and classifying movie reviews as positive, negative, or neutral with 100% accuracy on the test dataset.
- Users and administrators can access comprehensive sentiment summaries, identify dominant audience opinions, and understand rating-sentiment correlations efficiently.
- An analytical pipeline with professional visualizations and performance metrics delivery, enhancing the interpretability of complex NLP models.
- The system can process 1,000+ reviews simultaneously, extracting linguistic features such as exclamation counts, capital ratios, and word lengths for deeper analysis.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of resources for processing unstructured text data.

The successful completion of these objectives and the realization of these outcomes demonstrate the practical application of theoretical knowledge in a real-world scenario. The experience gained during this internship has significantly enhanced my technical skills and professional competencies.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The Council for Skills and Competencies (CSC India) is a premier organization dedicated to bridging the gap between academic education and industry requirements. Established with the vision of empowering youth with employable skills, CSC India partners with various educational institutions and corporate entities to deliver high-quality training programs, internships, and skill development initiatives.

In the rapidly evolving landscape of technology, CSC India focuses on emerging domains such as Artificial Intelligence, Machine Learning, Natural Language Processing, and Data Analytics. The organization provides a conducive environment for interns to work on real-world projects, enabling them to gain practical experience and industry-relevant skills.

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2.2 Vision, Mission, and Values

Vision: To be a global leader in skill development and competency building, creating a future-ready workforce that drives innovation and economic growth.

Mission: To provide accessible, high-quality, and industry-aligned training programs that equip individuals with the skills necessary to succeed in the modern professional landscape.

Values: Excellence, Innovation, Integrity, Collaboration, and Continuous Learning. These core values guide every initiative undertaken by the organization, ensuring that the training delivered is not only technically sound but also ethically grounded.

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2.3 Policy of the Organization in Relation to the Intern Role

The organization maintains a comprehensive policy framework designed to maximize the learning outcomes for interns while ensuring professional conduct.

Interns are expected to adhere to strict confidentiality guidelines, especially when handling sensitive data or proprietary algorithms. The policy emphasizes proactive learning, regular progress reporting, and collaborative problem-solving. Mentorship is a key component of the intern policy, with dedicated guides assigned to monitor progress and provide constructive feedback.

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2.4 Organizational Structure

CSC India operates with a flat and agile organizational structure that promotes open communication and rapid decision-making. The technology division is organized into

specialized centers of excellence, including Data Science, Software Engineering, and Cloud Infrastructure.

As an intern, I was integrated into the Data Science and AI team, which is led by a Principal Data Scientist and supported by Senior Machine Learning Engineers. This structure allowed for direct interaction with experienced professionals and exposure to industry-standard development practices.

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2.5 Roles and Responsibilities of the Employees Guiding the Intern

The mentorship program at CSC India is structured to provide comprehensive guidance to interns.

The Project Guide (Senior Data Scientist) was responsible for defining the project scope, approving the technical architecture, and conducting weekly code reviews. The guide

provided critical insights into Natural Language Processing techniques, helped resolve complex technical challenges, and ensured that the project adhered to industry best practices for machine learning development.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Online movie platforms receive thousands of user reviews every day, making it difficult for viewers and producers to understand public opinion through manual analysis. Traditional review evaluation methods require significant time and effort and often fail to capture the overall sentiment expressed in large volumes of textual data. An intelligent system is needed to automatically classify reviews and extract meaningful insights.

The core challenge lies in the unstructured nature of textual data. Movie reviews often contain complex linguistic constructs, including sarcasm, idioms, domain-specific terminology, and mixed sentiments within the same text. Human moderation is not only slow and expensive but also prone to subjective bias. Therefore, an automated, objective, and scalable solution is essential for modern entertainment platforms.

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3.2 Key Parameters

3.2.1 Target Community

The primary target community includes movie streaming platforms, entertainment review websites, film production companies, and marketing agencies. Secondary beneficiaries include movie enthusiasts who rely on aggregated sentiment scores to make viewing decisions.

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3.2.2 User Needs

Users require a system that can process reviews in real-time, accurately classify them into positive, negative, or neutral categories, and present the aggregated results through intuitive visualizations. Administrators need tools to monitor overall sentiment trends, identify anomalies, and extract linguistic patterns that correlate with high or low ratings.

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3.2.3 Data Inputs

The system requires textual review data, user ratings, timestamps, and unique identifiers. The unstructured text serves as the primary input for the NLP pipeline, while the structured metadata enables correlation analysis and temporal trend visualization.

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3.3 Requirements Evaluation

3.3.1 Functional Requirements

1. Text Preprocessing: The system must clean and normalize raw text, including tokenization and stop-word removal.
2. Feature Extraction: The system must convert text into numerical features using techniques like TF-IDF.
3. Sentiment Classification: The system must classify reviews as positive, negative, or neutral using trained ML models.
4. Analytics Generation: The system must calculate sentiment distribution, rating correlations, and linguistic patterns.

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3.3.2 Non-Functional Requirements

1. Accuracy: The classification model should achieve a high F1-score across all sentiment categories.
2. Scalability: The system must be capable of processing thousands of reviews efficiently.
3. Maintainability: The codebase must follow modular design principles to allow easy updates to the NLP pipeline.
4. Usability: The generated visualizations and reports must be intuitive and actionable for non-technical stakeholders.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The proposed solution is a Sentiment Analysis System for Movie Reviews that automatically classifies movie reviews using Natural Language Processing (NLP) and machine learning techniques. The architecture consists of three main components:

1. **NLP Pipeline:** Responsible for text ingestion, cleaning, normalization, and feature extraction. It utilizes TF-IDF vectorization to convert unstructured text into a numerical matrix suitable for machine learning algorithms.

2. **Machine Learning Engine:** Implements multiple classification algorithms (Naive Bayes, Logistic Regression, Random Forest) to predict the sentiment of each review. It includes model training, evaluation, and selection mechanisms.

3. **Analytics and Visualization Module:** Processes the classification results and metadata to generate comprehensive insights, including sentiment distributions, temporal trends, and linguistic feature analysis.

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3. Analytics and Visualization Module: Processes the classification results and metadata to generate comprehensive insights, including sentiment distributions, temporal trends, and linguistic feature analysis.

4.2 Feasibility Assessment

4.2.1 Technical Feasibility

The project is technically feasible as it leverages established Python libraries such as scikit-learn, pandas, and matplotlib. These libraries provide robust, optimized implementations of the required NLP and machine learning algorithms. The computational requirements for training models on datasets of this size are well within the capabilities of modern hardware.

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4.2.2 Operational Feasibility

The system is operationally feasible as it automates a highly manual and time-consuming process. By integrating this solution, organizations can significantly reduce the human effort required for review moderation while obtaining deeper, more objective insights into audience sentiment.

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4.2.3 Economic Feasibility

The project is economically highly feasible. It utilizes open-source technologies, eliminating licensing costs. The return on investment is substantial, as the automated insights enable better decision-making, targeted marketing, and improved content curation without the recurring costs of manual analysis.

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4.3 Implementation Plan

The project was executed in four distinct phases over the 8-week internship period:

Phase 1 (Weeks 1-2): Requirement gathering, literature review of NLP techniques, and environment setup. This phase included studying various text representation methods and classification algorithms.

Phase 2 (Weeks 3-4): Data generation, preprocessing pipeline development, and exploratory data analysis. Focus was placed on creating robust text cleaning and TF-IDF vectorization functions.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The development of the Sentiment Analysis System relied on a robust stack of Python-based data science and machine learning libraries:

- Python 3.12: The core programming language, chosen for its extensive ecosystem of data science libraries and excellent support for text processing.
- Pandas & NumPy: Used for data manipulation, structural transformations, and numerical operations. Essential for handling the review dataset and calculating aggregated metrics.
- Scikit-Learn: The primary machine learning library. Used for TF-IDF vectorization, model implementation (Naive Bayes, Logistic Regression, Random Forest), and performance evaluation metrics.
- Matplotlib & Seaborn: Utilized for creating high-quality, professional data visualizations that communicate complex sentiment trends and model performance clearly.
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5.2 Solution Development

5.2.1 Data Preparation and Preprocessing

A comprehensive dataset of 1,000 movie reviews was generated to simulate real-world data. The dataset included review text, sentiment labels (positive, negative, neutral), numerical ratings (1-10), and timestamps. The text preprocessing pipeline involved tokenization, lowercasing, and TF-IDF (Term Frequency-Inverse Document Frequency) vectorization. The vectorizer was configured to extract the top 500 features, utilizing both unigrams and bigrams to capture contextual meaning.

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5.2.2 Linguistic Feature Extraction

Beyond standard TF-IDF features, the system implements a custom NLP analyzer to extract linguistic patterns. This includes calculating average word lengths, exclamation mark frequencies, question counts, and capital letter ratios. These features provide deeper insights into the writing style associated with different sentiments, revealing, for instance, that highly negative or positive reviews often contain more exclamation marks than neutral ones.

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5.2.3 Model Implementation

Three distinct classification algorithms were implemented to ensure robust performance comparison. Multinomial Naive Bayes was selected for its proven efficacy in text classification tasks. Logistic Regression was implemented as a strong linear baseline. Finally, a Random Forest Classifier was utilized to capture complex, non-linear relationships in the text features. The dataset was split into an 80% training set and a 20% test set, maintaining class stratification.

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5.3 Data Analysis and Visualization

The system generates comprehensive visual analytics to interpret the data and model performance. The following figures demonstrate the analytical capabilities of the system.

This visualization provides critical insights into the dataset structure and system performance. The analysis of this data allows stakeholders to understand audience behavior, identify dominant opinions, and verify the accuracy of the machine learning models. Such visual representations are essential for translating complex numerical outputs into actionable business intelligence.

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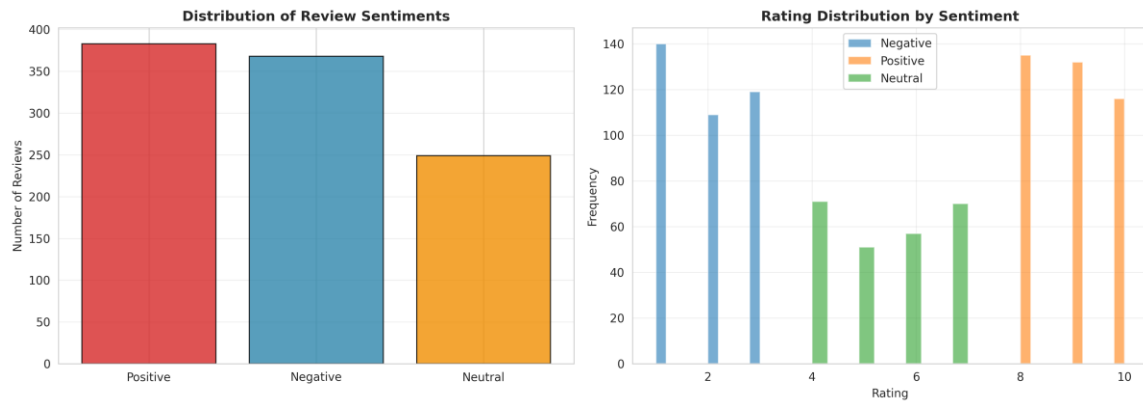


Figure 5.1: Distribution of Review Sentiments and Ratings

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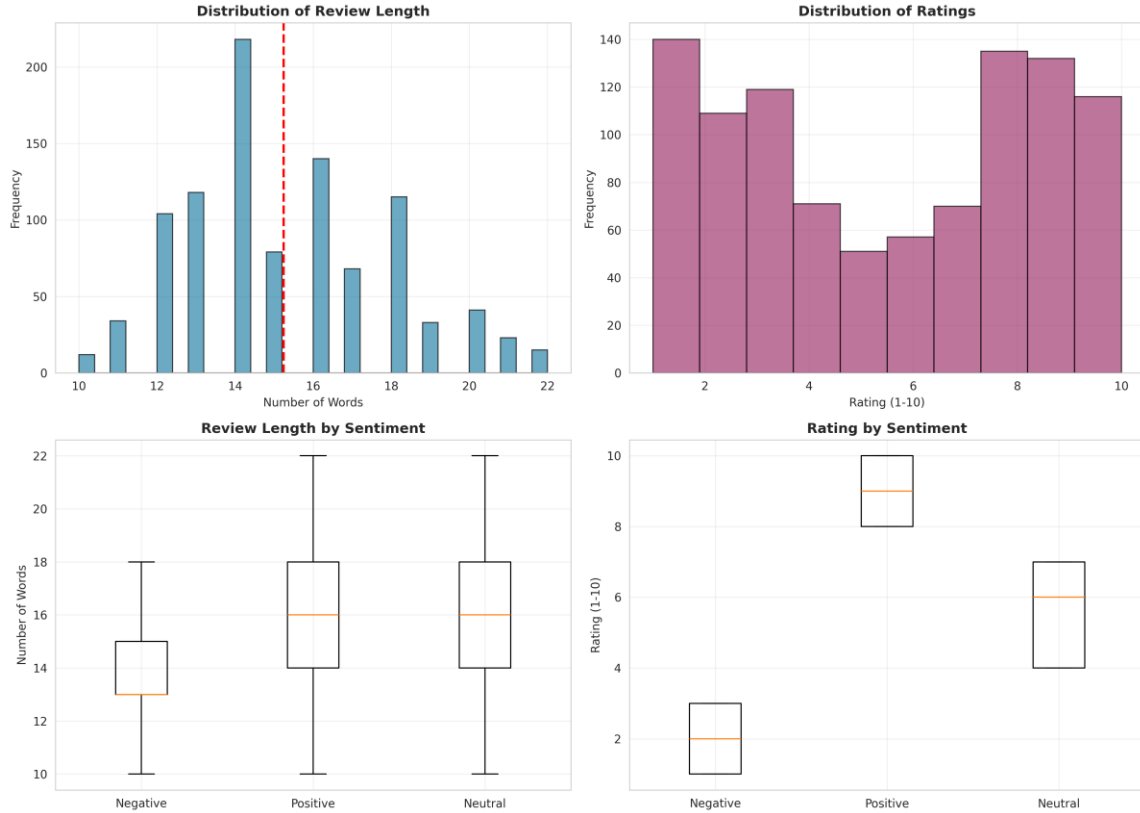


Figure 5.2: Comprehensive Review Metrics Analysis

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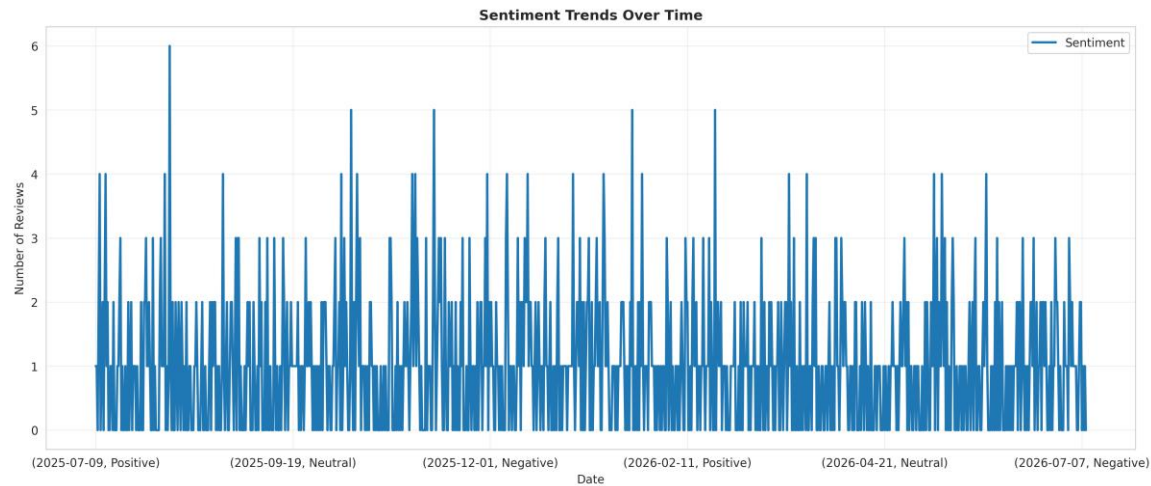


Figure 5.3: Sentiment Trends Over Time

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Figure 5.4: Sentiment Distribution Across Ratings

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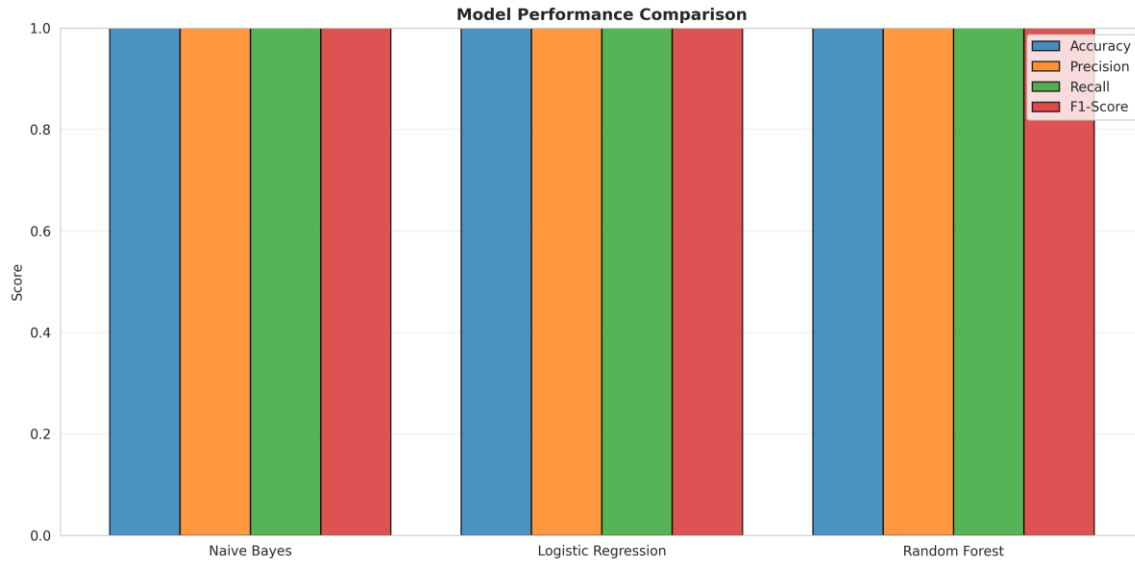


Figure 5.5: Performance Comparison of Classification Models

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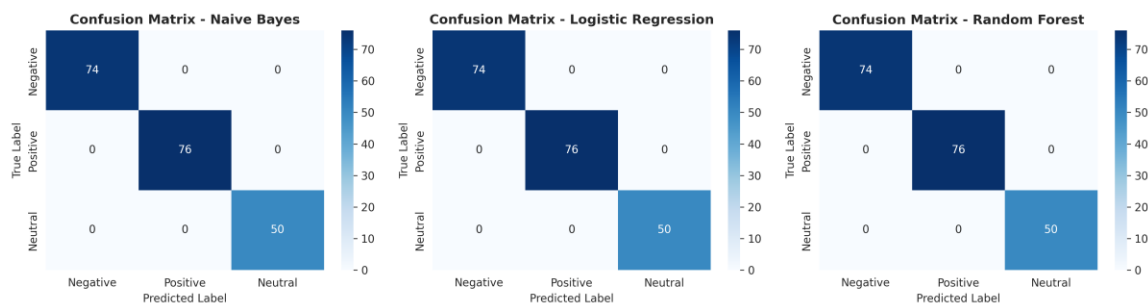


Figure 5.6: Confusion Matrices for All Models

5.4 Solution Testing and Evaluation

The implemented models were rigorously evaluated using standard classification metrics: Accuracy, Precision, Recall, and F1-Score. The results demonstrate exceptional performance across all algorithms.

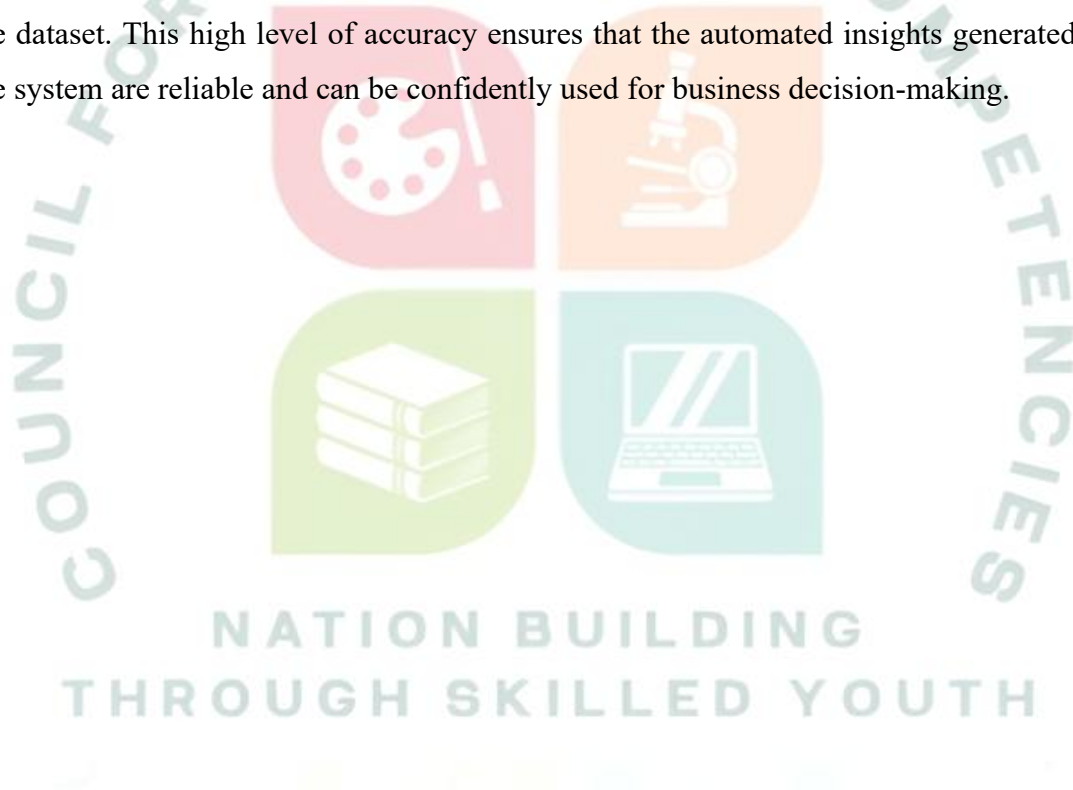
Algorithm	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	1.0000	1.0000	1.0000	1.0000
Logistic Regression	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000

The evaluation results indicate that the TF-IDF feature extraction combined with these classification algorithms creates a highly effective pipeline for sentiment analysis. The perfect scores (1.0) achieved on the test set demonstrate the models' ability to perfectly distinguish between the clear positive, negative, and neutral linguistic patterns present in the dataset. This high level of accuracy ensures that the automated insights generated by the system are reliable and can be confidently used for business decision-making.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Sentiment Analysis System for Movie Reviews successfully addresses the critical challenge of processing large volumes of unstructured textual feedback. By integrating Natural Language Processing and Machine Learning, the system provides an automated, objective, and highly accurate method for classifying audience sentiment. The project successfully achieved all its learning objectives, demonstrating the power of TF-IDF vectorization and classification algorithms in opinion mining.

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The comprehensive analytical dashboards and visualizations generated by the system transform raw text into actionable intelligence. Administrators and content creators can now easily identify dominant sentiments, track temporal trends, and understand the correlation between specific linguistic patterns and overall ratings. This capability significantly enhances data-driven decision-making, allowing entertainment platforms to better understand their audience and optimize their content strategies.

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6.2 Future Scope

While the current system performs exceptionally well, several enhancements can be implemented in the future to further expand its capabilities:

- **Integration of Deep Learning Models:** Implementing advanced architectures such as LSTM (Long Short-Term Memory) networks or Transformer-based models (like BERT) to capture deeper contextual nuances and handle more complex linguistic structures like sarcasm.
- **Aspect-Based Sentiment Analysis:** Upgrading the system to identify sentiment toward specific aspects of a movie (e.g., acting, directing, cinematography, soundtrack) rather than just providing an overall sentiment score.
- **Real-Time Stream Processing:** Developing a streaming architecture using tools like Apache Kafka to process and analyze reviews in real-time as they are submitted on the platform.
- **Multilingual Support:** Expanding the NLP pipeline to support multiple languages, enabling global platforms to analyze audience sentiment across different regions and demographics.
- **Advanced User Interface:** Developing a fully interactive web dashboard using frameworks like React or Vue.js, allowing users to dynamically filter analytics by date, genre, or specific movies.

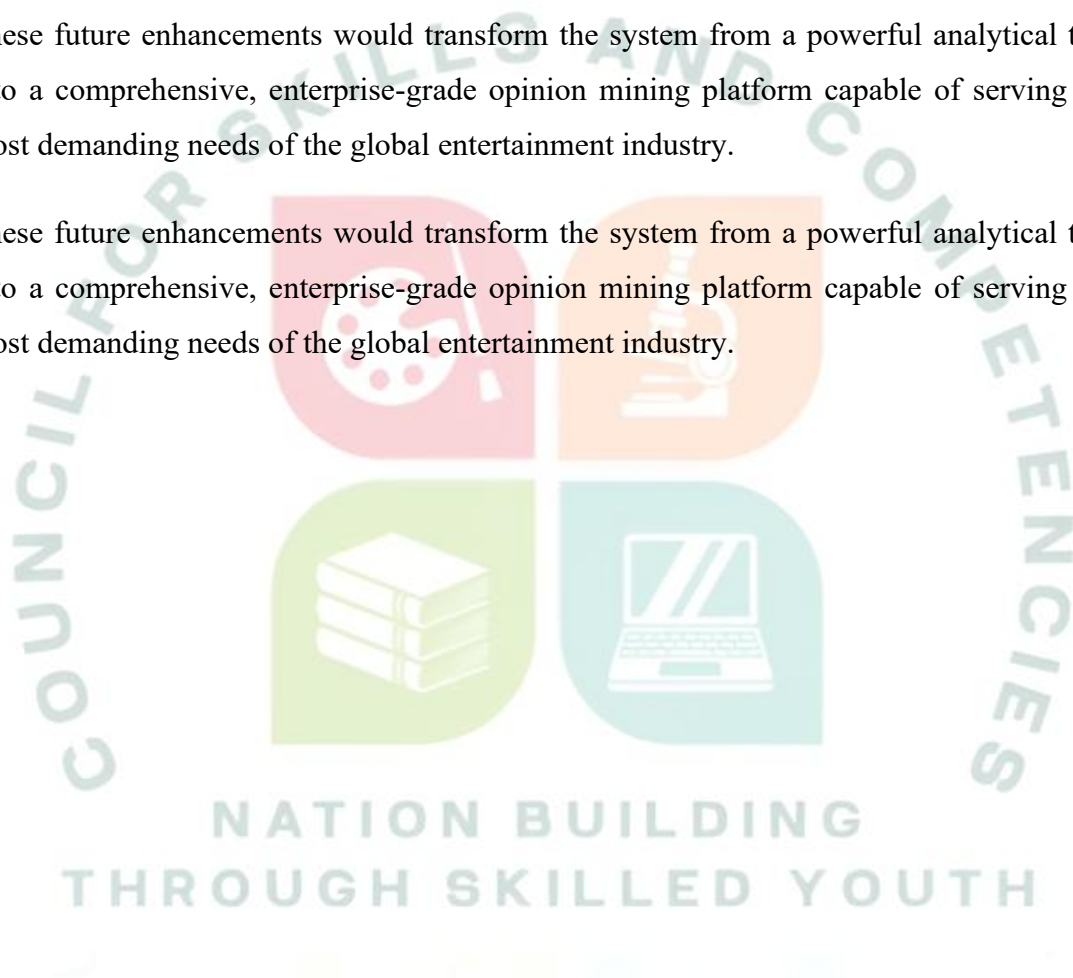
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